

Video Transcript

Video 2: The Role of ETL in Data Engineering (05:24)

When it comes to leveraging data, one of the terms that you'll hear is ETL, extract, transform, and load. And if we were to look at the hierarchy of data needs, the one that was introduced by Monica Rogati, which is a pretty good visualization of the stages that one has to go through in order to be able to derive insights from data. You can see there that we go through collection, through storage, transformations, labeling, optimization, all the way into machine learning and AI. Now when it comes to ETL, we're looking at the bottom layers of that pyramid. You can see there that there's quite a bit going on, instrumentation, logging, user generated content, etc., throughout those first tiers.

Now, the diagram that you'll often see looks something like the following. You have a number of sources, and these could be sensors out in the field. They could be different types of data generation; they could be web pages that you're scraping, think of Google and others, and all of this information is being brought back. That is, you are extracting it from all of those sources out in the field and then you have to do some transformation, and once you give it some shape, you push it into some type of data store or databases. Now, this seems like a very easy picture to put your arms around and that you know, you're picking up different parts from different places, you're bringing them together, and pushing them potentially into some tables.

Well, as it turns out, the picture is much broader. And as you can see here, ETL sits right in the middle between collection, storage, and processing, and analysis, and consuming that data. And because this is a much broader picture, it's worth going through some of these verticals. And I'll start off here with collection. You can see there that simply by looking at those course blocks and those of course are simplifications already, we're looking at three different sections there of types of data. You have application data that is coming off of desktop application, it's coming off of mobile devices, it's coming off of web applications. Then you have another category there that would be monitoring, logging, migration of data. And if you look at the picture right next to it where you have records and then files, you can see that the type of data, the granularity, the way in which it is typically generated is different between these two. And then, there at the bottom,

you have internet of things, you have sensors, you have IoT platforms. And there you have streaming data, a great volume and velocity typically. And you could have the variety as well.

But in general, you can see here that although we say collection easily, there is quite a bit to account for. You have different formats, you have different sources, you have different speeds, you have different granularity. And so, simply that is in itself a challenge. Then you have the next tier, which is the vertical tier of storage, and you can have everything from databases that are running on the cloud to your traditional relational database management system in all the different types of data stores that we have today, the key value pair stores, we have the document databases, we have the scalable, distributed, decentralized, and all of this richness would fall into that when it comes to what's coming off of the logging and monitoring.

We typically have flat files there that are going into file systems. And then, at the very end where we're talking about streaming, we have a different type of database yet that can handle those type of loads and that type of stream of information. And so, that brings us there to the middle, the last two columns, which would be process analysis and consumption of that, are ones that are more upstream from ETL. So, I will not talk about them in detail, but you can see there that this is a very rich and vast environment when it comes to the type of transformations of extractions of loading that needs to happen. And that because of it, it is a wonderfully creative set of technologies that are addressing these needs. And at the same time, they're a great deal that can be derived when it comes to value from this data that is moving through these pipelines. So, with that in mind, we'll go ahead and talk about ETL, we'll talk about a specific set of open-source tools that we can use, but I wanted to show you this to begin with simply for you to understand the full picture when it comes to data science and data engineering.